Broc Weselmann

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Education

University of Central Florida — Orlando, FL
B.S. in Computer Science — *Aug. 2024 – May 2027*

Projects

Encyclopedia AI | *Go, JavaScript, Ollama, OpenRouter, SQLite, GitHub Actions*

- Automated article writing by architecting a **9-stage multi-agent Go pipeline** (intake, outline, section drafting, evaluate, compare, plan, revise, metadata) that critiques and revises its own output
- Kept tens-of-minutes reasoning runs reliable by building a provider layer for Ollama and OpenAI-compatible APIs with structured-output repair and retries, plus **resumable SQLite-backed jobs** streamed over SSE

SINEBot | *JavaScript, Node.js, discord.js, Chart.js, SQLite, GitHub Actions, DigitalOcean*

- Built a Discord bot for UCF's Society of Innovation and Neurodiversity in Engineering used daily by **about a dozen members on each of 2 servers**, by creating a plugin registry of daily games with Chart.js analytics
- **Increased engagement and retention** and sparked friendly competition by shipping leaderboards, trend graphs, and distributions from user feedback

Trip Advisor | *TypeScript, React, Express, MongoDB, JWT, GitHub Actions*

- Eliminated manual FTP re-uploads of the whole project to the server (a few minutes per deploy, per developer) for a **team of 6** by automating DigitalOcean deploys with a **GitHub Actions CI/CD pipeline** and PR integration tests
- Secured user accounts for a React/TypeScript web app by setting up the repo, MongoDB database, JWT auth, and bcrypt password hashing

Drone Route Optimizer | *Python, OR-Tools, NumPy, Streamlit, Folium, Plotly*

- **Routed a drone across a graph of 2,500+ nodes** for NextEra Energy's KnightHacks VIII challenge by designing the routing algorithm with VRP heuristics and OR-Tools

GPT-1 Language Model | *Python, PyTorch (ROCm), NumPy*

- Generated Shakespeare-style prose over a 1,000-token context window by pre-training a **GPT-1-like transformer in PyTorch** with scaled residual projections and checkpointing

A Frail Qubit | *GDScript, Godot Engine*

- **Built a story-based puzzle game** with **over 60 handcrafted levels** by developing all mechanics, narrative, and pixel art solo in Godot with GDScript

The New Order: The Last Days of Europe | *Paradox Script, GitLab*

- **Coordinated a fully remote team of 50+ volunteers** behind a total-overhaul mod with **500,000+ players** via GitLab
- **Onboarded over a dozen volunteers** by authoring the Git guide they used, while implementing event chains, GUIs, and content in Paradox Script

Experience

Grocery Clerk — Aug. 2021 – Present
Publix

- **Earned 2 promotions** (Front Service Clerk to Cashier to Grocery Clerk) by cross-training at two stores
- Kept shelves stocked and aisles maintained while averaging **30 hours per week** during school, by unloading shipments and restocking

Technical Skills

Languages: Python, Go, JavaScript, TypeScript, C, C++, GDScript
Frameworks & Data: React, Node.js, Express, PyTorch, NumPy, SQLite, MongoDB, MySQL
Agentic AI: Claude Code, Codex, multi-agent LLM pipelines, Ollama, Generative AI, Transformers
Tools & Platforms: Git, GitHub Actions, GNU/Linux, DigitalOcean, Godot Engine